Notes: Berger, Dew-Becker & Giglio (RES, 2020)

Uncertainty Shocks as Second-Moment News Shocks

Contents

1 Big picture

- **Question.** Do forward-looking uncertainty shocks cause recessions, or are downturns mostly associated with the realization of large shocks today?
- **Key distinction.** Realized volatility is the size of shocks that just occurred; uncertainty is the expected variance of future shocks.
- **Main result.** Innovations in realized stock-market volatility are followed by contractions in employment and industrial production. Pure second-moment news shocks are not.
- **Risk-premium result.** Investors historically paid large premia to hedge realized volatility shocks but little or no premium to hedge forward-looking uncertainty shocks.
- **Economic takeaway.** Aggregate volatility matters, but the contractionary component is realized volatility associated with bad news today, not uncertainty about future volatility by itself.

2 Data and measurement

2.1 Realized volatility

Realized volatility, rv_t, is constructed from daily S&P 500 returns and aggregated to the monthly frequency. It captures the size of shocks that have already occurred.

Detailed interpretation: Realized volatility is not a belief about future risk. It is a measurement of current shocks.

2.2 Option-implied volatility

The benchmark forward-looking measure is one-month option-implied volatility $v_{1,t}$ from S&P 500 options. Robustness checks use six-month implied volatility.

Detailed interpretation: Implied volatility contains news about future variance, but it must be decomposed because it is correlated with realized volatility.

2.3 Macroeconomic variables

The benchmark VAR uses rv_t, $v_{1,t}$, the federal funds rate, log industrial production, and log employment from 1983–2014.

Detailed interpretation: Employment and IP are the real-activity outcomes used to test whether volatility and uncertainty shocks are contractionary.

3 Models and empirical specifications

3.1 VAR state vector

$$Y_t = [rv_t, v_{1,t}, FFR_t, \log IP_t, \log EMP_t]'.$$

Detailed interpretation: Realized volatility is ordered before implied volatility so that the uncertainty news shock is orthogonal to current realized volatility.

3.2 Second-moment news shock

$$News_t = E_t \left[\sum_{j=1}^{n} rv_{t+j} \right] - E_{t-1} \left[\sum_{j=1}^{n} rv_{t+j} \right].$$

The uncertainty shock is the component of this news orthogonal to the innovation in rv_t.

Detailed interpretation: This is the paper's core identification idea: uncertainty is news about future realized volatility, not volatility realized today.

3.3 Scaling

The benchmark sets $n = 24$ months and scales the realized-volatility and news shocks so that they have equal cumulative effects on expected future volatility.

Detailed interpretation: The shocks are comparable because they have equal future-uncertainty content.

4 Identification and empirical design

The central identification problem is that realized volatility and expected future volatility are correlated. The paper assumes that pure uncertainty news has no contemporaneous effect on realized volatility on average, while realized volatility can affect expected future volatility. This separates current large shocks from second-moment news about future shocks. Robustness checks weaken this timing assumption, use daily data to address time aggregation, use alternative uncertainty measures, and test model-based simulations.

5 Section-by-section study guide

- **Introduction.** Distinguishes realized volatility from uncertainty and previews the result.
- **Identification.** Adapts news-shock methods to second moments.
- **VAR results.** Shows that rv shocks are contractionary and uncertainty/news shocks are not.
- **Robustness.** Shows results survive alternative measures, orderings, and estimation methods.
- **Pricing.** Shows investors pay to hedge realized volatility but not pure future volatility news.
- **Model.** Shows left-skewed shocks can rationalize the findings.

6 Tables and figures

6.1 Table 1: Relationship between employment, IP, and volatility

Read: Realized volatility drives out implied volatility when both enter regressions for employment and IP growth.

Detailed interpretation: Table 1 shows why a decomposition is needed: implied volatility looks contractionary partly because it is correlated with realized volatility.

Economic message: Current large shocks, not pure uncertainty news, drive the negative relation with real activity.

TABLE 1

Relationship between employment, IP, and volatility

	Employment		Industrial production	
v_1	−0.10**	0.05	−0.09*	0.16
	(0.04)	(0.07)	(0.05)	(0.12)
rv		−0.14**		−0.23*
		(0.07)		(0.13)

Note: Results from regressions of changes in log employment and IP on the current value and four lags of implied (v_1)

Table 1: Relationship between employment, IP, and volatility

BERGER *ET AL.* UNCERTAINTY SHOCKS 49

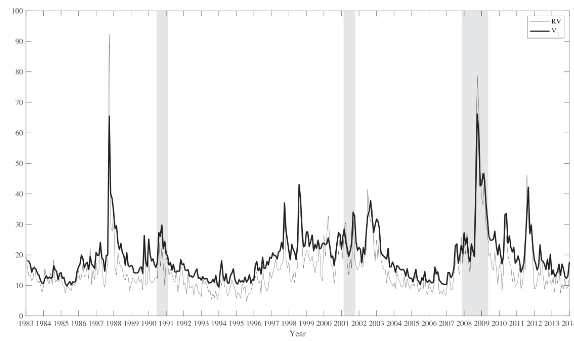


FIGURE 1

Time series of realized volatility and expectations.

Note: Time series of realized volatility (RV), and 1-month option-implied volatility (V_1), in annualized standard deviation units. Grey bars

Figure 1: Time series of realized volatility and expectations

6.2 Figure 1: Time series of realized volatility and expectations

Read: Realized and implied volatility both spike in crisis periods and recessions.

Detailed interpretation: The figure motivates the identification problem: the two series are related but conceptually different.

Economic message: Raw volatility spikes mix current bad news with future uncertainty.

6.3 Figure 2: IRFs from benchmark VAR

Read: Realized volatility shocks lower employment and IP; news shocks do not.

Detailed interpretation: The shocks have equal future uncertainty content, so the difference comes from current realized volatility.

Economic message: Realized volatility is the contractionary object.

6.4 Figure 3: Forecast error variance decomposition

Read: News shocks explain meaningful uncertainty variation but little real-activity variation.

Detailed interpretation: News shocks are not irrelevant; they are just not contractionary.

Economic message: Moving uncertainty is not the same as moving output.

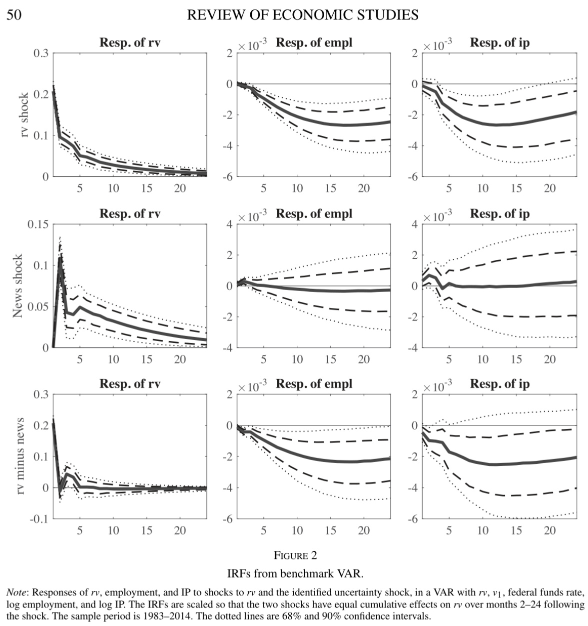


Figure 2: IRFs from benchmark VAR

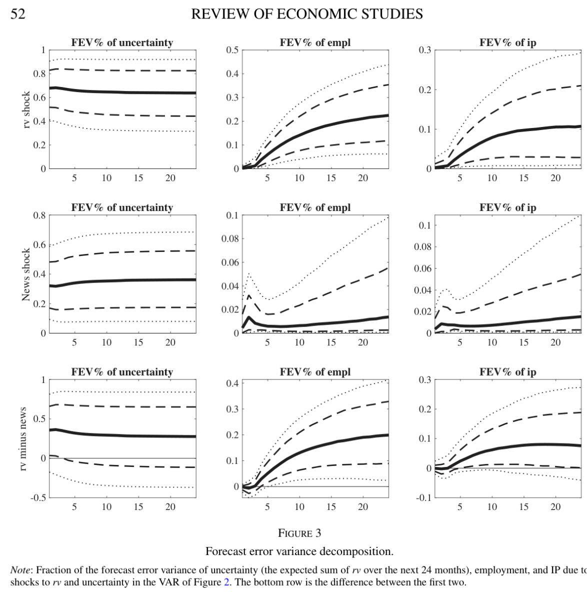


Figure 3: Forecast error variance decomposition

6.5 Figure 4: Fitted uncertainty

Read: Fitted uncertainty can be decomposed into rv-driven and news-driven components.

Detailed interpretation: The news component lines up with plausibly uncertainty-related events such as policy debates and the euro crisis.

Economic message: The identified news shocks are interpretable but not recessionary.

BERGER *ET AL.* UNCERTAINTY SHOCKS 53

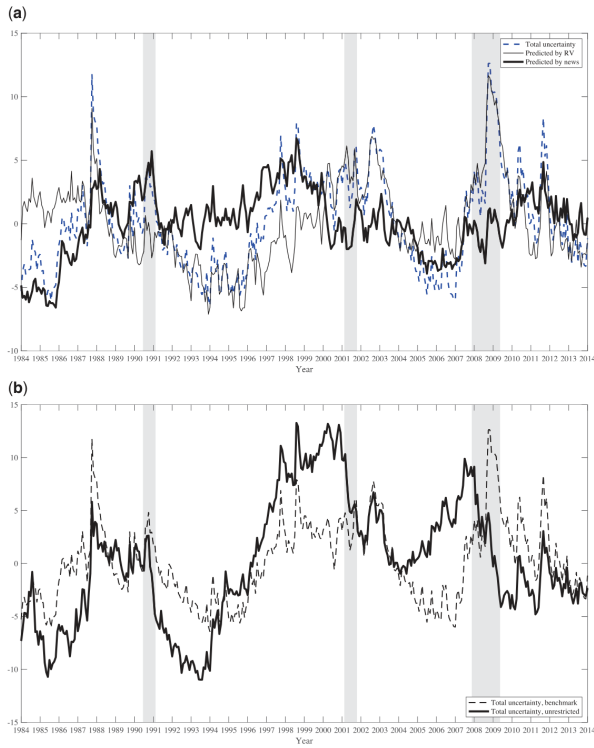


FIGURE 4

Fitted uncertainty. (a) Decomposition of total uncertainty in the benchmark specification; (b) Total uncertainty in benchmark versus unrestricted specification.

Note: The top panel reports a decomposition of total uncertainty (the conditional expectation of the sum of *rv* over the next 24 months)

Figure 4: Fitted uncertainty

6.6 Tables 2 and 3: Validation and volatility forecasting

Read: Table 2 links VAR-implied uncertainty to alternative uncertainty indexes. Table 3 shows implied volatility predicts future realized volatility.

Detailed interpretation: Option-implied volatility has useful information, but it must be separated from realized volatility.

Economic message: Markets forecast future volatility, but news about future volatility is not necessarily contractionary.

TABLE 2

VAR-implied and alternative measures of uncertainty

	RV	News	p(difference)
BBD	14.6***	7.1	0.28
	(2.3)	(6.8)	
BBD (monetary)	23.2***	10.5	0.20
	(7.1)	(8.9)	
BBD (stock market)	38.7***	19.7	0.26
	(14.4)	(18.1)	
Michigan	0.79*	0.82	0.96
	(0.44)	(0.59)	

Note: The table shows regressions of alternative measures of uncertainty on the components of VAR-implied uncertainty from *rv* and news shocks. BBD is the Bloom *et al.* (2015) measure, BBD (monetary) and BBD (stock market) are the corresponding subcomponents of that measure, and Michigan is a measure of uncertainty from the Michigan Consumer Survey (used by Leduc and Liu (2016)). The last column reports the *p*-value for a test of the difference between the two coefficients.

(a) Table 2

TABLE 3

Predictability of 6-month rv

Predictors	(1)	(2)	(3)	(4)	(5)
rv_t	0.16*	0.15	0.16*	0.13	0.09
	(0.10)	(0.09)	(0.09)	(0.09)	(0.09)
$v_{1,t}$	0.57***	0.46***	0.60**	0.58***	0.60***
	(0.14)	(0.16)	(0.29)	(0.15)	(0.15)
$v_{6,t}$			−0.03		
			(0.30)		
rv_{t-1}		0.12**			
		(0.05)			
FFR				0.01	
				(0.01)	
$\Delta empl$				−4.98	
				(10.87)	
Δip				−4.81	
				(3.57)	
PC1					−0.010
					(0.009)
PC2					−0.017
					(0.012)
PC3					−0.011
					(0.007)
$R_{S\&P}$					0.12
					(0.41)
Default spread					0.07
					(0.09)
Adj. R^2	0.44	0.45	0.44	0.45	0.46

Note: Results of linear predictive regressions of realized volatility over the next 6 months on lagged *rv*, option-implied volatility, and various macroeconomic variables, with Hansen–Hodrick (1980) standard errors using a 6-month lag window. PC1–3 are principal components from the data set used in Ludvigson and Ng (2007). The default spread is the difference in yields on Baa and Aaa bonds. The sample is 1983–2014.

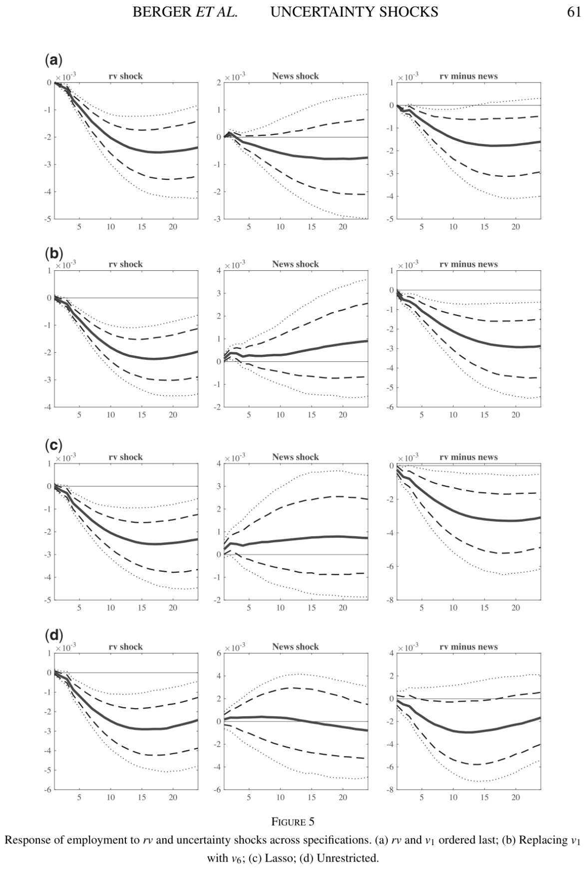
(b) Table 3

6.7 Figures 5 and 6: Robustness across specifications

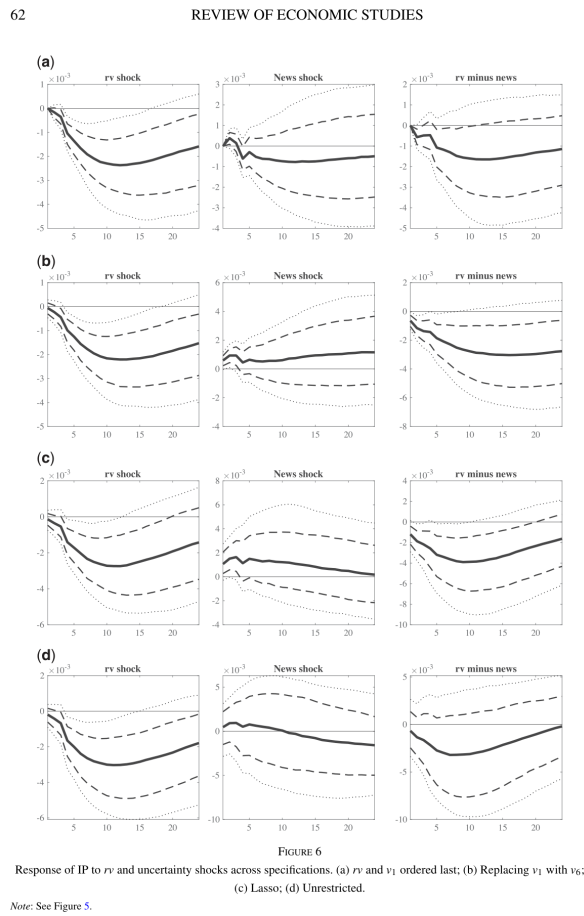
Read: Employment and IP responses show the same pattern across alternative orderings, v_6, lasso, and unrestricted specifications.

Detailed interpretation: The key result is robust to major specification changes.

Economic message: The result is not a VAR-ordering artifact.



(a) Figure 5: Employment



(b) Figure 6: IP

6.8 Figure 7: Forward variance claims

Read: Short-term realized variance claims have high Sharpe ratios; longer-horizon forward variance claims have near-zero premia.

Detailed interpretation: Investors pay to hedge realized volatility but not pure uncertainty news.

Economic message: Asset prices confirm the VAR interpretation.

6.9 Table 4: Skewness

Read: Most macro variables have negative skewness.

Detailed interpretation: Left skewness makes large realized volatility more likely to coincide with bad first-moment shocks.

Economic message: Realized volatility can be contractionary because it reveals bad news.

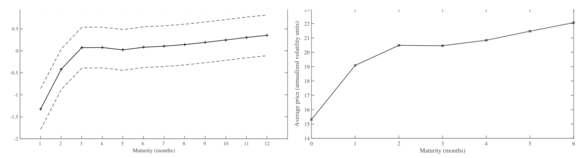


FIGURE 7

Forward variance claims: returns and prices. (a) Sharpe ratios; (b) Average prices.

Note: Panel A shows the annualized Sharpe ratio for the forward variance claims, constructed using variance swaps. The returns are calculated assuming that the investment in an n-month variance claim is rolled over each month. Dotted lines represent 95% confidence intervals. All tests for the difference in Sharpe ratio between the 1-month variance swap and any other maturity confirm that they are statistically different with a p-value of 0.03 (for the second month) and <0.01 (for all other maturities). The sample used is 1996–2013. For more information on the data sources, see Dew-Becker *et al.* (2017). Panel B shows the average prices across maturities of synthetic forward variance claims constructed from option prices for the period 1983–2014. All prices are reported in annualized standard deviation units. Maturity zero corresponds to average realized volatility.

Figure 7: Forward variance claims: returns and prices

TABLE 4
Skewness

	Skewness		Kelley Skewness		
Panel A: real economic activity	Monthly	Quarterly	Monthly	Quarterly	Sample
Employment	-0.31	-0.87^{***}	-0.14^{***}	-0.23^{***}	1952–2016
	(0.33)	(0.19)	(0.04)	(0.07)	
Capacity utilization	-1.06^{***}	-1.20^{***}	-0.03	-0.15	1969–2016
	(0.31)	(0.37)	(0.05)	(0.11)	
IP	0.14	-0.63^{**}	-0.05	-0.09	1952–2016
	(0.50)	(0.30)	(0.04)	(0.08)	
IP, starting 1960	-0.87^{**}	-0.99^{**}	-0.07	-0.16^{**}	1960–2016
	(0.32)	(0.35)	(0.04)	(0.08)	
Output		-0.32		0.02	1952–2016
		(0.24)		(0.07)	
Non-durable consumption		-0.33		0.09	1952–2016
		(0.34)		(0.08)	
Durable consumption		-0.25		-0.14^{**}	1952–2016
		(0.43)		(0.06)	
Investment		-0.62^{**}		0.03	1952–2016
		(0.26)		(0.10)	
Residential investment		-0.41		-0.13^{*}	1952–2016
		(0.29)		(0.07)	
Non-residential investment		-0.76^{**}		-0.21^{***}	1952–2016
		(0.32)		(0.06)	

Panel B: skewness of S&P 500 monthly returns	Estimate	Std. Err.
Implied (since 1990)	-1.90^{***}	(0.09)
Realized (since 1926)	0.19	(0.54)
Realized (since 1948)	-0.54^{***}	(0.20)
Realized (since 1990)	-0.67^{***}	(0.20)

Note: Panel A reports the skewness of log growth rates of measures of real activity. The first column reports the skewness of monthly changes, the second column the skewness of quarterly changes. Kelley skewness is computed as $KSK = \frac{P(90)+P(10)-2P(50)}{P(90)-P(10)} \in [-1,1]$ where $P(*)$ denotes percentiles. Panel B reports the realized skewness of S&P 500 monthly returns in different periods, as well as the implied skewness computed by the CBOE using option prices. Standard errors are constructed by bootstrapping with 1,000 replications.

Table 4: Skewness

6.10 Table 5, Figure 8, and Figure 9: Structural model

Read: The left-skewed model matches skewness, variance-risk premia, and VAR impulse responses.

Detailed interpretation: The model rationalizes why rv shocks are bad while uncertainty news shocks are not.

Economic message: Left-skewed fundamental shocks link realized volatility to recessions.

TABLE 5
Model calibration

Panel A: Moments	Model			Data		
	Mean	Std.	Skewness	Mean	Std.	Skewness
Output	1.99	1.75	-3.02	1.33	1.92	-0.11
Investment	1.99	2.87	-4.27	2.35	7.43	-0.03
Consumption	1.99	1.46	-4.83	1.20	1.06	-0.28
Returns	5.92	15.00	-3.30	7.46	14.77	-0.48
VIX	19.57	1.20	1.75	20.05	7.75	1.94

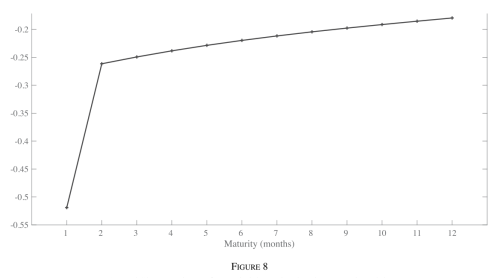
Panel B: Corr. of VAR and structural shocks		Structural shocks		
		Jv_t	η_t	ε_t
	RV	0.79	0.00	0.07
VAR identified shocks	Uncertainty	0.24	0.87	-0.02

Note: Panel A reports the mean, standard deviation, and skewness of financial and macroeconomic variables in the data and in the model (returns are reported in annualized terms). Panel B shows the correlation between the structural shocks in the model and the shocks identified in the VAR.

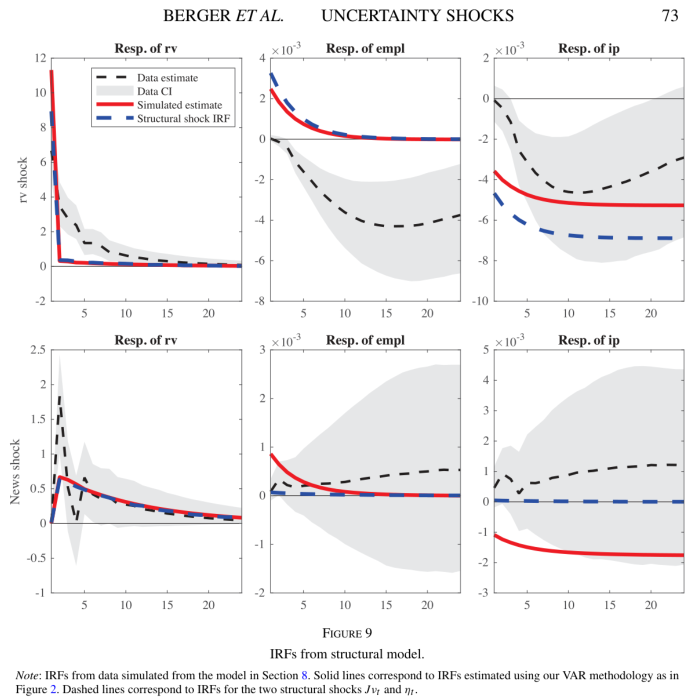
dynamics but also for asset prices and realized volatility. Integrals are calculated using Gaussian quadrature with twenty points. Euler equation errors are less than $10^{-5.0}$ across the range of the state space that the simulation explores and have an average absolute value of $10^{-5.3}$. The use of a global solution method allows for high accuracy in the solution, but also makes it infeasible to search over many parameters or estimate the model, which is why we explore just a single calibration here.

8.4. *Simulation results*

Table 5: Model calibration



(a) Figure 8



(b) Figure 9

7 Short summary

- The paper separates realized volatility from forward-looking uncertainty.
- Realized volatility shocks are followed by contractions in employment and IP.
- Pure second-moment news shocks predict future volatility but do not significantly reduce real activity.
- Investors pay large premia to hedge realized volatility, but not longer-horizon uncertainty news.
- Negative skewness explains why realized volatility is associated with downturns.

8 Conclusion

8.1 Core conclusion

The macroeconomic effects often attributed to uncertainty shocks are mostly associated with realized volatility shocks.

8.2 Economic intuition

A spike in realized volatility often reflects a large negative shock that has already occurred. Pure uncertainty news changes beliefs about future risk without necessarily delivering bad news today.

8.3 Contribution

The paper clarifies the identification problem in the uncertainty literature by separating current volatility from second-moment news.

8.4 Asset-pricing interpretation

Variance-claim returns show that realized volatility is a bad state investors pay to hedge, while pure future uncertainty news has little average premium.

8.5 Limitations

The rv shock is an empirical object and not a single structural primitive. It may reflect many underlying disturbances.

8.6 Takeaway

Uncertainty and volatility are not the same. Realized volatility is historically contractionary; pure uncertainty news is not.